

Colon & Colorectal Cancer — Biomarker Testing Intelligence

Alert Window: Last 30 Days · Generated: 2026-04-12 11:46:48

Situation Summary

Stage IV: 2 of 18
Biomarker Count: 13 entries
Top Targets: ctDNA (4) · MSI/MMR (4) · TP53 (2)
Breakthrough: 6 entries scored ≥ 4

Research & Guidelines Timeline

Biomarker testing entries ranked by significance score. Higher scores indicate greater clinical relevance. Always discuss with your oncologist.

MEDIUM	Stage IV	BROAD SCOPE	VERIFIED	GENOMIC PROFILING	PUBLISHED MAR 28, 2026	9
The Evolving Role for Repeat Molecular Testing in Metastatic Colorectal Cancer						
This study evaluates the evolving role for repeat molecular testing in metastatic colorectal cancer.						
pubmed.ncbi.nlm.nih.gov						
HIGH	All Stages	ctDNA	VERIFIED	LIQUID BIOPSY	PUBLISHED APR 03, 2026	7
Liquid biopsies using circulating tumor DNA for surveillance of gastrointestinal cancers in Hispanics: first real-world data report						
This study evaluates liquid biopsies using circulating tumor DNA for surveillance of gastrointestinal cancers in Hispanics.						
pubmed.ncbi.nlm.nih.gov						
HIGH	All Stages	BROAD SCOPE	VERIFIED	LIQUID BIOPSY	PUBLISHED MAR 30, 2026	6
Utility of Urinary Extracellular Vesicles for Colorectal Cancer: Comparison of DNA Quality and MRD Detection						
This study evaluates the utility of urinary extracellular vesicles for colorectal cancer through a comparison of DNA quality and MRD detection.						
pubmed.ncbi.nlm.nih.gov						
HIGH	All Stages	ctDNA	VERIFIED	LIQUID BIOPSY	PUBLISHED MAR 14, 2026	6
Multistep ctDNA Monitoring of Minimal Residual Disease in Colorectal Cancer Liver Metastases: From Tissue NGS to Highly Sensitive Digital PCR Platforms						
This study evaluates multistep ctDNA monitoring of minimal residual disease in colorectal cancer liver metastases, from tissue NGS to highly sensitive digital PCR platforms.						
pubmed.ncbi.nlm.nih.gov						

HIGH

Early Detection

ctDNA

VERIFIED

LIQUID BIOPSY

PUBLISHED MAR 25, 2026

⚡ 4

Blood-based circulating tumour DNA (ctDNA) tests for colorectal cancer screening: Systematic review and meta-analysis of diagnostic accuracy

This study evaluates blood-based circulating tumour DNA (ctDNA) tests for colorectal cancer screening through a systematic review and meta-analysis of diagnostic accuracy.

pubmed.ncbi.nlm.nih.gov

HIGH

Early Detection

ctDNA

VERIFIED

LIQUID BIOPSY

PUBLISHED APR 01, 2026

⚡ 3

Circulating Tumor Cells as the Liquid Biopsy Foray into Noninvasive Colorectal Cancer Screening

This study evaluates circulating tumor cells as the liquid biopsy foray into noninvasive colorectal cancer screening.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

MSI/MMR

VERIFIED

GENOMIC PROFILING

PUBLISHED MAR 14, 2026

⚡ 3

Dual testing with immunohistochemistry and PCR for enhanced accuracy in dMMR/MSI detection in Greek colorectal cancer population: the impact of retesting

This study evaluates dual testing with immunohistochemistry and PCR for enhanced accuracy in dMMR/MSI detection in the Greek colorectal cancer population and the impact of retesting.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

MSI/MMR

VERIFIED

RESEARCH

PUBLISHED MAR 18, 2026

⚡ 2

Molecular Profiling of Germline Variants in the DNA Mismatch Repair Genes in Chinese Colorectal Cancer Patients

This study evaluates molecular profiling of germline variants in the DNA mismatch repair genes in Chinese colorectal cancer patients.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

Genomic Profiling

VERIFIED

GENOMIC PROFILING

PUBLISHED MAR 17, 2026

⚡ 2

A paradigm shift in genetic predisposition to colorectal cancer: the impact of germline multigene panel testing on diagnosis and management

This study evaluates the impact of germline multigene panel testing on diagnosis and management in the context of genetic predisposition to colorectal cancer.

pubmed.ncbi.nlm.nih.gov

MEDIUM

Early Detection

BROAD SCOPE

VERIFIED

GENOMIC PROFILING

PUBLISHED MAR 27, 2026

⚡ 1

Biomarkers in Colorectal Cancer: Clinically Relevant Diagnostic and Prognostic Molecular Features, and the Future of Precision Medicine

This study evaluates biomarkers in colorectal cancer, focusing on clinically relevant diagnostic and prognostic molecular features, and the future of precision medicine.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

MSI/MMR

VERIFIED

RESEARCH

PUBLISHED MAR 24, 2026

⚡ 1

dMMR prediction from colorectal cancer histopathology: Leveraging non-tumor and low-magnification regions

This study evaluates dMMR prediction from colorectal cancer histopathology by leveraging non-tumor and low-magnification regions.

MEDIUM

All Stages

EGFR

VERIFIED

RESEARCH

PUBLISHED MAR 23, 2026

⚡ 1

Circulating Butyrate Attenuates Cetuximab Efficacy in Colorectal Cancer Through EGFR and AMPK-Wip1 Signaling

This study evaluates how circulating butyrate attenuates cetuximab efficacy in colorectal cancer through EGFR and AMPK-Wip1 signaling.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

TP53

VERIFIED

GENOMIC PROFILING

PUBLISHED MAR 13, 2026

⚡ 1

TP53 Loss Fuels mTORC1 Activation and Autophagy Suppression to Drive Immune-Cold Colorectal Cancer

This study evaluates how TP53 loss fuels mTORC1 activation and autophagy suppression to drive immune-cold colorectal cancer.

pubmed.ncbi.nlm.nih.gov

Appendix A — Patient-Friendly Summary

Plain-language overview for patients, caregivers, and family members

What's happening

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. The results determine which FDA-approved treatments may be available to you. The entries below represent the highest-scoring study per biomarker target drawn from the full research archive — not limited to the current report window. Some entries may predate the timeline above; they are included because they remain the most clinically significant reference available for that biomarker.

ctDNA

Circulating Tumor DNA as a Biomarker for Recurrence After Curative Resection of Colorectal Cancer Liver Metastases: A Systematic Review and Meta-Analysis

This study evaluates circulating tumor DNA as a biomarker for recurrence after curative resection of colorectal cancer liver metastases through a systematic review and meta-analysis.

TP53

Clinical impact of TP53 mutation status on survival outcomes in metastatic colorectal cancer

This study evaluates the clinical impact of TP53 mutation status on survival outcomes in metastatic colorectal cancer.

BRAF

Detection of KRAS, NRAS and BRAF Mutations in Liquid Biopsy from Patients with Colorectal Cancer

This study evaluates the detection of KRAS, NRAS, and BRAF mutations in liquid biopsy from patients with colorectal cancer.

NGS

Exploratory biomarker findings from regorafenib plus toripalimab in patients with refractory metastatic colorectal cancer (REGOTORI study)

This study evaluates exploratory biomarker findings from regorafenib plus toripalimab in patients with refractory metastatic colorectal cancer (REGOTORI study).

MSI/MMR

Prognostic Impact of Blood Tumor Mutational Burden in pMMR/MSS Metastatic Colorectal Cancer Assessed by FoundationOne() Liquid CDx

This study evaluates the prognostic impact of blood tumor mutational burden in pMMR/MSS metastatic colorectal cancer assessed by FoundationOne Liquid CDx.

RAS

Effect of sex differences on the emergence of ctDNA RAS mutations in RAS wild-type colorectal cancer

This study evaluates the effect of sex differences on the emergence of ctDNA RAS mutations in RAS wild-type colorectal cancer.

Genomic Profiling

Integrated Molecular Profiling of Colorectal Cancer by Tumor Location: Evidence from a Real-World Cohort with Primary and Metastatic Samples

This study evaluates integrated molecular profiling of colorectal cancer by tumor location using evidence from a real-world cohort with primary and metastatic samples.

HER2

Targeting HER2 in Metastatic Colorectal Cancer: Current Therapies, Biomarker Refinement, and Emerging Strategies

This study evaluates targeting HER2 in metastatic colorectal cancer, focusing on current therapies, biomarker refinement, and emerging strategies.

KRAS

Next-Generation Sequencing-Based Detection of KRAS G12D Variants in Colorectal Cancer: A Retrospective Cohort Study

This study evaluates Next-Generation Sequencing-Based Detection of KRAS G12D Variants in Colorectal Cancer through a Retrospective Cohort Study.

EGFR

Circulating Butyrate Attenuates Cetuximab Efficacy in Colorectal Cancer Through EGFR and AMPK-Wip1 Signaling

This study evaluates how circulating butyrate attenuates cetuximab efficacy in colorectal cancer through EGFR and AMPK-Wip1 signaling.

miRNA

Diagnostic Performance of Circulating miRNA-92a and Related Circulating miRNA Panels for Colorectal Cancer: An Updated Systematic Review and Meta-Analysis

This study evaluates the diagnostic performance of circulating miRNA-92a and related circulating miRNA panels for colorectal cancer through an updated systematic review and meta-analysis.

IHC

Integration of deep learning and radiomic features from multiplex immunohistochemistry images for reproducible Multi-Outcome prediction in a Multi-Center study of colorectal cancer

This study evaluates the integration of deep learning and radiomic features from multiplex immunohistochemistry images for reproducible multi-outcome prediction in a multi-center study of colorectal cancer.

Stool DNA

An algorithm-enhanced stool DNA system improves the differential diagnosis of colorectal cancer versus Crohn's disease in high-risk symptomatic patients

This study evaluates an algorithm-enhanced stool DNA system that improves the differential diagnosis of colorectal cancer versus Crohn's disease in high-risk symptomatic patients.

What this means for you

If you have been diagnosed with metastatic colorectal cancer, ask your oncologist: *"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"* Testing should happen at diagnosis — before treatment begins — because results take 1–2 weeks and can shape your first treatment decision. Ask specifically about KRAS, NRAS, BRAF V600E, MSI/MMR status, HER2, NTRK, and TMB.

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This is not medical advice. Always speak with your oncologist before making any treatment decisions.

Appendix B — Colon & Colorectal Cancer: Biomarker Testing Reference

What Is Biomarker Testing?

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. A single next-generation sequencing (NGS) panel can check many gene mutations at once. Results typically take 1–2 weeks and should be obtained at diagnosis, before treatment begins.

Key Biomarkers — Test All mCRC at Diagnosis

RAS (KRAS/NRAS) · BRAF V600E · MSI/MMR status · HER2 · NTRK fusion · TMB

Testing Methods

NGS (next-generation sequencing) — checks many mutations at once from tumor tissue

Liquid biopsy / ctDNA — blood test that detects tumor DNA; useful when tissue is unavailable

IHC (immunohistochemistry) — used to test MMR status and HER2

PCR — used for specific mutation detection (e.g. MSI, KRAS)

Questions to Ask Your Oncologist

"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"

"Was my sample taken from the primary tumor or a metastatic site — and does it need to be retested?"

"How long will results take, and will they affect my first treatment decision?"

"Is a liquid biopsy appropriate for my situation?"

Abbreviations

BRAF V600E — a specific gene mutation found in ~8–10% of colorectal cancers

ctDNA — circulating tumor DNA; tumor DNA fragments found in the bloodstream

dMMR — mismatch repair deficient; indicates the tumor may respond to immunotherapy

HER2 — a protein that drives growth in ~2–3% of colorectal cancers

IHC — immunohistochemistry; a lab staining method used to detect proteins in tissue

KRAS G12C — a specific gene mutation found in ~3–4% of colorectal cancers

mCRC — metastatic colorectal cancer (Stage IV)

MSI-H — microsatellite instability-high; indicates the tumor may respond to immunotherapy

NGS — next-generation sequencing; a method that checks many gene mutations at once

NTRK — a rare gene fusion found in <1% of colorectal cancers

RAS — a gene family (KRAS/NRAS) whose mutation status affects treatment selection

TMB — tumor mutational burden; a measure of how many mutations are in a tumor

Key Resources

ClinicalTrials.gov · NCCN Guidelines (nccn.org) · Colorectal Cancer Alliance (ccalliance.org) · Fight Colorectal Cancer (fightcrc.org) · ACS (cancer.org)

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About This Report

This report is generated automatically from publicly available medical databases including PubMed, ASCO Journal of Clinical Oncology, Annals of Oncology, and NCI Cancer.gov. Entries are classified using deterministic rule-based logic and scored by clinical significance. AI-generated summaries are grounded exclusively in the source data provided.

Limitations

This report does not represent a complete picture of all published research. Database coverage, feed availability, and relevance filtering affect what appears. Research findings may not yet reflect current clinical guidelines. Always verify information at the original source before making clinical decisions.

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https://petroai1492.github.io/delta/timelines/coloncancer_biomarker_timeline.pdf